

Can He

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EDUCATION

Northeastern University - Boston Campus , Boston, MA <i>M.S. in Computer Science, GPA: 3.94 / 4.0</i> Coursework: Computer Systems, Algorithms, Object-Oriented Design, Discrete Structures, Foundations of AI, Machine Learning	Sep 2023 - Apr 2026
Northeastern University - Boston Campus , Boston, MA <i>M.S. in Project Management, GPA: 3.761 / 4.0</i>	Sep 2013 - Jan 2016
Chongqing University of Technology , Chongqing, China <i>B.E. in Mechanical Engineering</i>	Sep 2008 - Jun 2012

TECHNICAL SKILLS

Languages: Python, Java, C, R, MQL4, SQL
AI / Computer Vision: OpenCV, PyTorch, Ultralytics YOLO, Scikit-Learn, NumPy, Pandas, image preprocessing, postprocessing
Backend / Data: RabbitMQ, AWS S3, SQLAlchemy, Alembic, pytest, DAOs, async upload queues, traceability metadata
DevOps / Observability: Docker, Kubernetes, Grafana, Prometheus, systemd, Git, CI-ready testing workflows, runtime health checks

WORK EXPERIENCE

Tristar AI , Boston, MA <i>Junior Software Developer</i> - Build production computer vision pipelines for manufacturing customers, translating inspection requirements into configurable capture, inference, traceability, and deployment workflows. - Develop Python services for image preprocessing/postprocessing, task/result schemas, RabbitMQ/S3 publishing, PLC rejection feedback, runtime health checks, and deployment configuration. - Improve reliability through pytest coverage, test doubles, monitoring hooks, Grafana/Prometheus metrics, Docker/Kubernetes deployment packaging, and deployment runbooks.	May 2026 - Present
Tristar AI , Boston, MA <i>Software Development Intern</i> - Implemented data-access objects, upload schedulers, camera/data capture utilities, defect-analysis scripts, configuration loaders, and local development test flows for customer-facing AI systems. - Validated behavior across camera, messaging, storage, and inference boundaries using pytest, dependency injection, local fixtures, and CI-ready project structure.	Jun 2025 - Jan 2026
Northeastern University , Boston, MA <i>Teaching Assistant - DS5220 Supervised Machine Learning</i> Supported ML course delivery by grading assignments, preparing answer keys, explaining supervised-learning concepts, and leading office hours and review sessions.	Sep 2024 - Jun 2025

SELECTED PROJECTS

Medegen Manufacturing Defect Detection Pipeline <i>Python, OpenCV, PyTorch, Ultralytics YOLO, RabbitMQ, AWS S3, PLC, systemd</i> - Owned end-to-end tub defect detection for four defect categories, spanning image preprocessing, model inference, postprocessing, classification, and line/camera-specific filtering. - Integrated capture, inference, rejection, results, station-status, alerting, and outbox modules in a six-camera, three-line production system with multiprocessing workers. - Enabled real-time quality feedback through FIFO-triggered capture, PLC-driven rejection, RabbitMQ/S3 publishing, Slack alerts, auditable records, and runtime failure-mode tests.
Amcor Bottle Traceability Pipeline <i>Python, OpenCV, Terpsichore, AWS S3, RabbitMQ, Prometheus</i> Built an end-to-end traceability pipeline connecting camera frames, station/team/factory metadata, task lifecycle states, result payloads, asynchronous upload queues, and monitored execution.
Greif Defect Data Analysis Pipeline <i>Python, Pandas, OpenCV, sensor/data streams</i> Analyzed defect records, image evidence, sensor/process streams, and timing signals to identify recurring failure patterns and support downstream inspection workflows.
Quantitative Trading & RL Systems <i>Python, MQL4, backtesting, MT4 gateway, reinforcement learning</i> Built XAUUSD/FX research systems with spread/slippage/commission-aware backtests, equity curves, risk metrics, trade ledgers, MT4 command gateways, optimization reports, and RL evaluation.

EARLIER ENGINEERING EXPERIENCE

Wuhu Automobile Advanced Technology Institute Co., Ltd. , Wuhu, Anhui, China <i>Project Manager</i> Led a 23-person engineering team developing lightweight carbon-fiber automotive components, achieving a 42% weight reduction on targeted Chery Tiggo 8 Pro PHEV components while managing testing, cost control, and production integration.	Sep 2020 - Sep 2023
Chery Holding Group Co., Ltd. , Wuhu, Anhui, China <i>Project Manager</i> Optimized chassis adaptation for fuel and hybrid vehicle lines through testing, model adjustment, technical data management, QC collaboration, patent support, and production guidance.	Sep 2016 - Sep 2020